

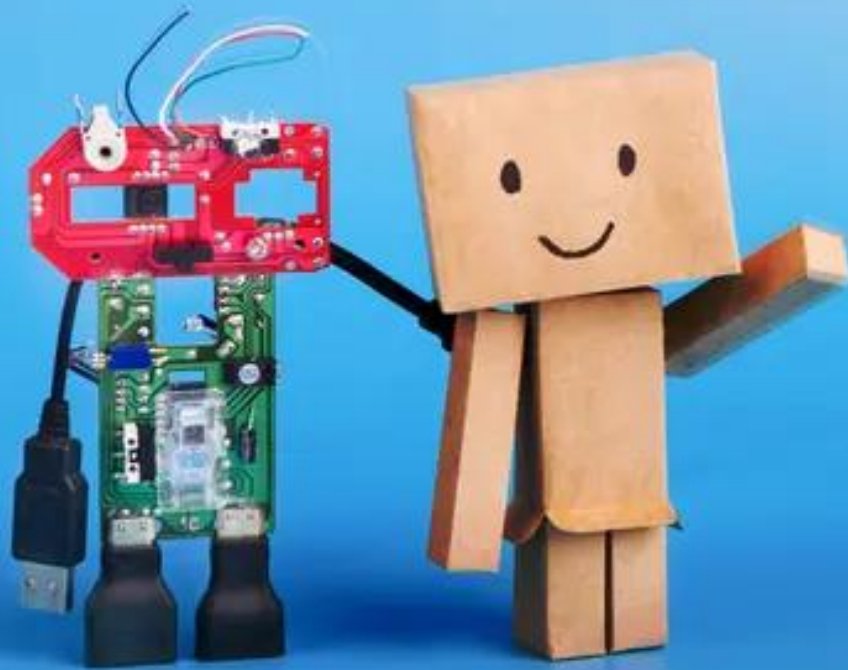
Lecture 1

AI-CR-503

Digital Technology Program

Madan Bhandari University of Science and Technology (MBUST)

UNIT 1 Introduction to People-Centered-AI



Artificial Intelligence (AI)

- Old definition in 1965.
 - “AI was described then as the ability of a machine or computer program to learn and think.”
- New definition in 2018.
 - “AI is defined as systems designed by humans that act in the physical or digital world by perceiving their environment, interpreting structured or unstructured data, reasoning on the knowledge derived from this data and deciding the best actions to achieve a given goal.”

People-centred AI (PCAI)

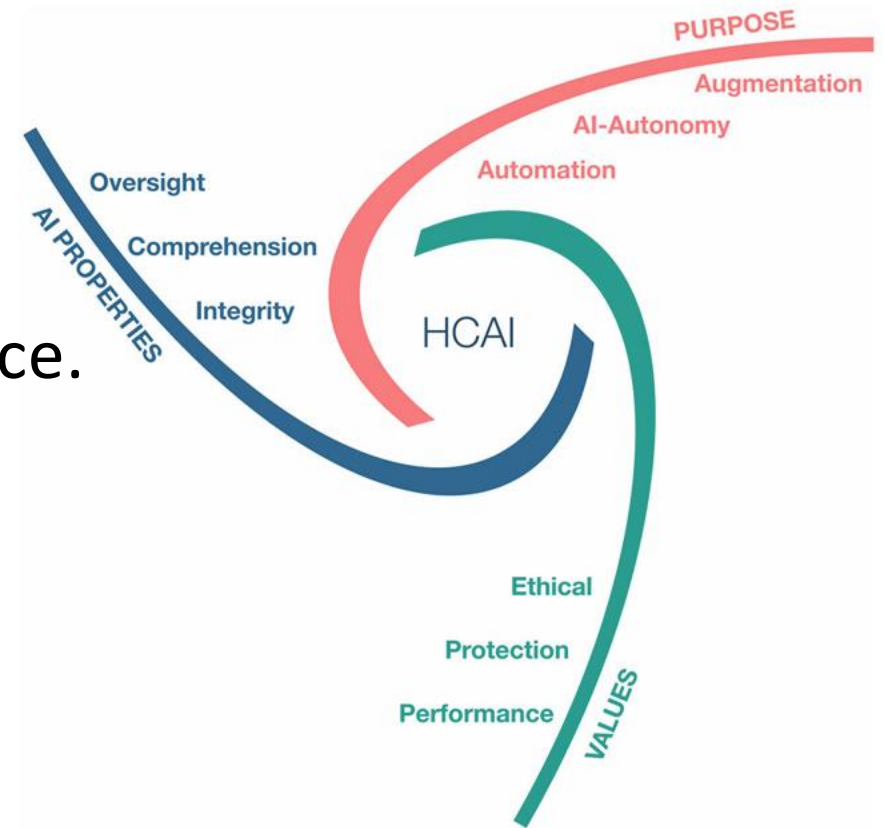
- People-centred AI (PCAI) is also known as Human-centred AI (HCAI)
- “People-Centered AI (HCAI) focuses on understanding purposes, human values, and desired AI properties in the creation of AI systems by applying Human Centered Design practices. HCAI seeks to augment human capabilities while maintaining human control over AI systems, by considering the necessity, context, and ethical and legal conditions of the AI system as well as promoting individual and societal well-being.”

Why we need People-centred AI (PCAI)?

- Growing use of AI have raised concern regarding risk and adverse consequences.
- Increasing interest in understanding impact of AI's on society.
- Aligning human values and needs in AI applications.

Elements of PCAI

1. Purpose: Augmentation, AI autonomy, and Automation.
2. Values: Ethics, protection, and performance.
3. Properties: Oversight, comprehension, and integrity.



1. Purpose

- It means considering the purpose of use of product, service, or systems.
 - It is an important element of people-centred AI system.
 - It includes augmentation, AI Autonomy, and Automation.
- a) Augmentation: It refers to the use of AI to enhance and amplify human capabilities, skills, and decision-making but without replacing human professionals.

- b) AI autonomy: It means AI has capacity to manifest agency via performing actions as evidenced by operating independently of humans to achieve goals.
- c) Automation: It refers to the undertaking of well-defined and repetitive tasks by AI without human intervention.

2. Values

- It is second element of people-centred AI approach.
- Values serve as a guiding framework for individuals and groups, to define what is important, just, and meaningful.
- It includes ethical, protection, and performance.

- a) Ethical: It includes dignity, Fairness and Justice.
- b) Protection: It includes privacy, safety, and protection from harm.
- c) Performance: It includes efficacy, effectiveness, and accuracy.

3. Properties

- The third element of people-centred AI system is properties and it includes oversight, comprehension, and integrity.
 - a) Oversight: It comprises accountability, controllability, responsibility, and sustainability.
 - b) Comprehension: Explainability, intelligibility, and transparency.
 - c) Integrity: Reliability, robustness, trustworthiness, and traceability.

Human-centred design: Process, methods, tools

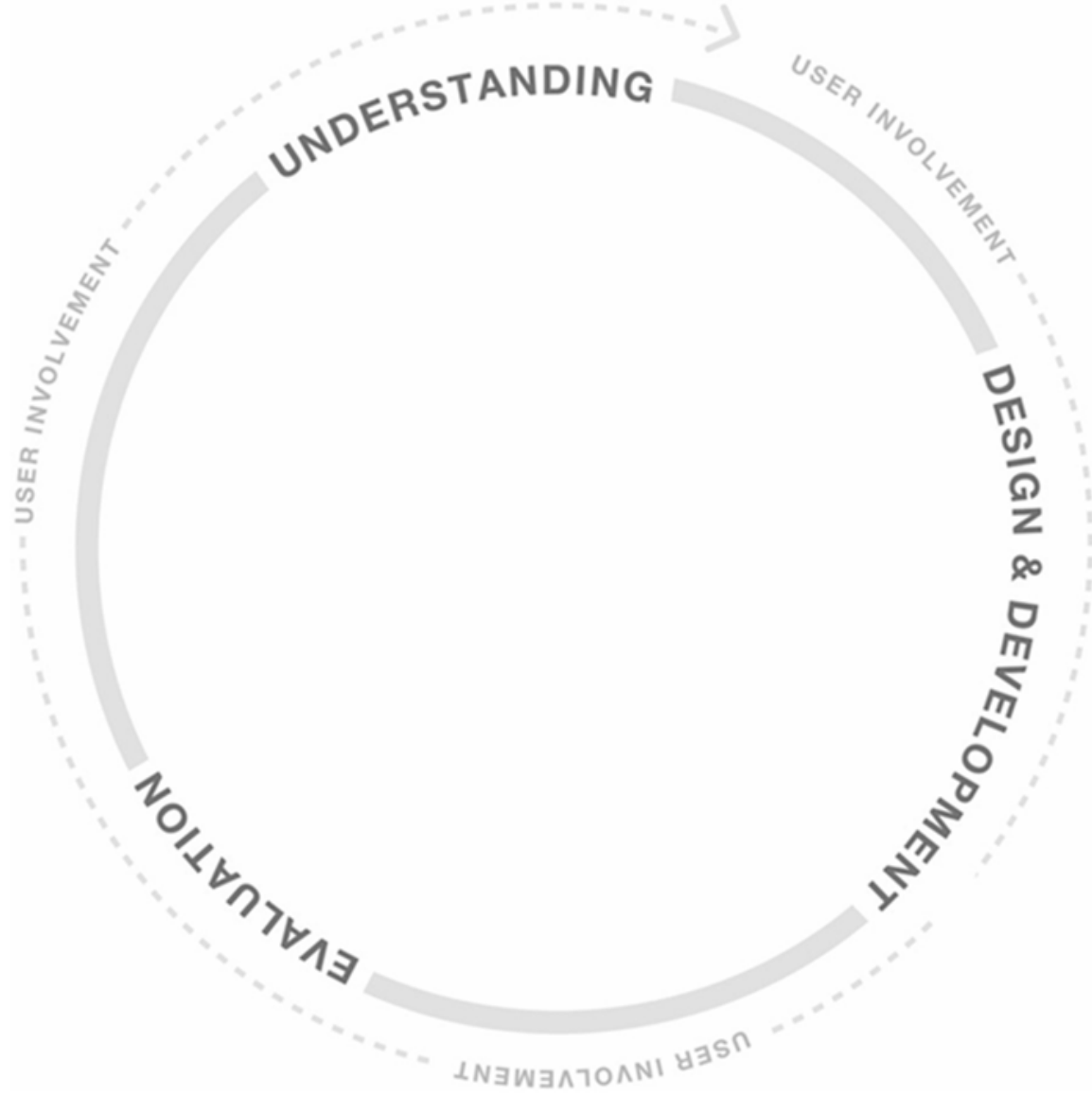
- Two new processes and methods being specifically developed for the design and development of people centred AI system are:
 1. User involvement
 2. Iterative process

1. User involvement

- It refers that people-centred design needs to be based upon an explicit understanding of users, tasks, and environments, i.e. the context of use.
- Products, systems, and services should be designed to take account of the people who will use them as well as other stakeholder groups, including those who will be directly or indirectly affected.

2. Iterative process

- Continuous iteration helps to progressively eliminate uncertainty during the development of people-centred AI systems.
- It implies that descriptions, specifications, and prototypes are constantly revised and refined with new information in order to minimize the risk of the system under development.



Research Agenda

- Theme 1: Purpose
- Theme 2: Values
- Theme 3: Properties
- Theme 4: Process, methods, and tools
- Theme 5: Human-centred design
- Theme 6: Conceptual clarity

Thank you.